

Toward Cooperative Driving Among Heterogeneous CAVs: A Safe Multi-Agent Reinforcement Learning Approach

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Abstract—With the advancement of Intelligent Transportation Systems and Vehicle-to-Everything communication technologies, the future traffic scenario is anticipated to be a mixed environment consisting of heterogeneous Connected and Autonomous Vehicles (CAVs) and Human-Driven Vehicles. Though reinforcement learning (RL) has shown tremendous potential in solving decision-making problems in autonomous driving, existing research often overlooks the heterogeneity of CAVs. Moreover, current RL methods lack robust safety guarantees, especially in mixed traffic scenarios where HDVs are often non-cooperative and uncontrollable. To address challenges of the cooperative driving among heterogeneous CAVs in dynamic and mixed traffic, we have developed a novel heterogeneous cluster safe algorithm and hybrid-critic training framework. Specifically, to overcome the limitations of existing methods primarily designed for homogeneous agents, we introduce a cluster policy-sharing mechanism. Additionally, we propose a safe policy update method with a high sample utilization rate, which combined with the cluster policy-sharing mechanism leads to the HCP4O algorithm. Furthermore, we have designed an innovative hybrid-critic training framework to enhance the safety driving performance of heterogeneous CAVs under various conditions. Finally, we train our model in hundreds of random and dynamic scenarios to validate the generalization ability of agents. The experimental results show that, compared to current advanced Multi-Agent RL algorithms and rule-based methods, our approach demonstrates significant advantages in safety and efficiency, especially in scenarios with a large number of agents.

Index Terms—Connected autonomous vehicles, heterogeneous and mixed traffic, multi-agent reinforcement learning, cluster policy-sharing mechanism, safe policy optimization.

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I. INTRODUCTION

A. Backgrounds and Motivations

IN RECENT years, though autonomous driving technology has received widespread attention, the safety issue remains a major obstacle to the development of autonomous driving [1]. Particularly in dynamic and complex traffic flows, fully autonomous vehicles tend to exhibit unstable and unsafe driving behaviors, which may lead to risks to personal safety and economic losses [2]. Fortunately, with significant progress in Vehicle-to-Everything (V2X) communication technologies, the integration of connected vehicle technology with autonomous driving is expected to overcome the limitations of information acquisition and collaboration capabilities, and thereby enhances the safety and efficiency of road traffic [3].

Utilizing V2X communication, Connected and Autonomous Vehicles (CAVs) can achieve cooperative driving in various traffic environments [4], [5]. The future connected traffic environment is supposed to be mixed traffic comprising of both heterogeneous CAVs and Human-Driven Vehicles (HDVs) [6], [7], as shown in Fig. 1. However, the decision-making process of HDVs might be non-cooperative and uncontrollable, which hence hinders the safety and efficiency of CAVs.

The decision-making module of CAVs, acting as their “intelligent brain,” determines the driving mode or behavior based on environmental information and vehicle status. There have been various approaches investigating this problem, such as rule-based methods [8], [9], [10], optimization-based methods [11], [12], [13], and learning-based methods [14], [15], [16]. Typically, rule-based methods prioritize road safety to avoid potential collisions with other vehicles regardless of the cost, while they are overly conservative, thereby leading to inefficient traffic flow. Optimization-based methods can provide precise solutions, with good interpretability and controllability. However, the scalability and efficiency issues remain in large-scale problems.

Learning-based methods can avoid the issues brought by the two aforementioned methods, not only enabling higher driving efficiency but also easier extension to multi-agent environments. In particular, reinforcement learning (RL) methods have demonstrated tremendous potential in addressing decision-making problems, achieving significant success in various difficult tasks [17], [18], [19]. There also exist some works extending single-agent RL algorithms to multi-agent reinforcement

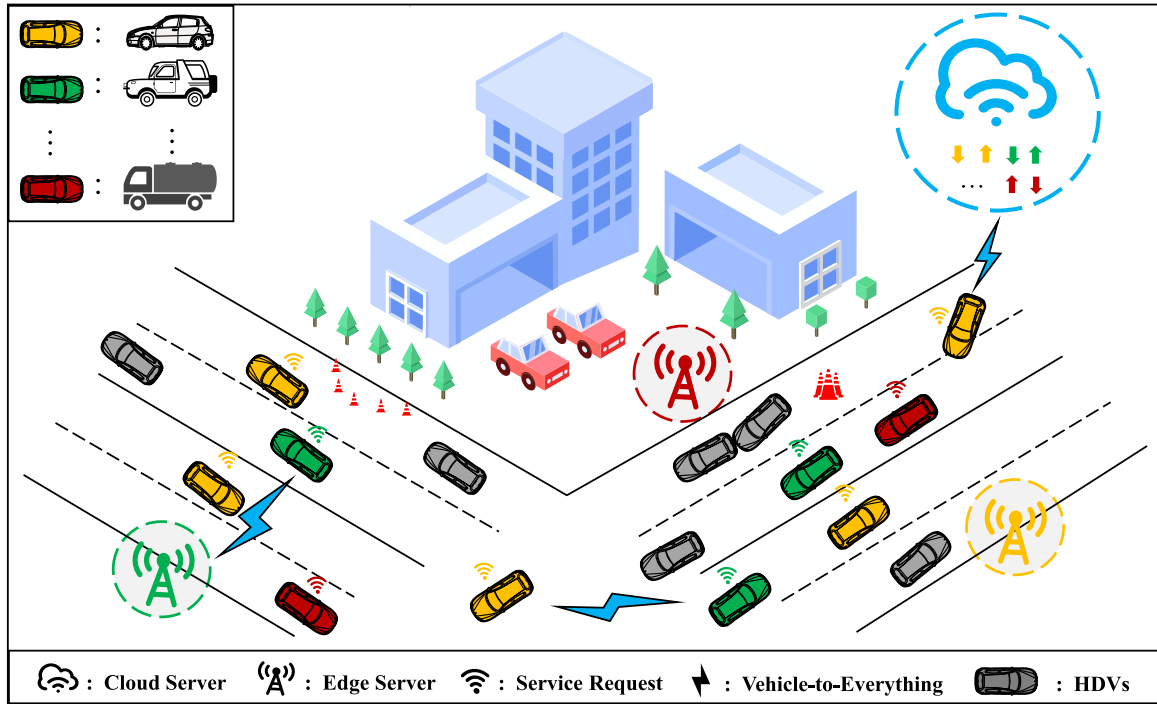


Fig. 1. A cooperative driving scenario involving heterogeneous CAVs, HDVs, and static obstacles.

learning (MARL) for addressing cooperative decision-making problems in CAVs [20], [21]. These studies tend to simplify CAVs as homogeneous agents and adopt parameter-sharing approaches, which however overlook the heterogeneity of CAVs, and hence cannot capture the diverse capabilities and behaviors of individual CAVs in real-world scenarios. Take sedans and trucks as a comparative example, sedans decelerate and stop quickly, but trucks require longer braking distances since they are heavier and have greater inertia. If a truck uses a braking strategy meant for sedans, it might not be able to avoid collisions. Besides, sedans can easily maneuver due to their smaller turning radius when turning, while trucks often cross lanes or struggle to make smooth turns due to their longer bodies. In addition, prior reinforcement learning methods (see e.g., [22], [23], [24]) lack robust safety guarantees. In certain scenarios, agents may prioritize maximizing rewards over ensuring safety, which however might potentially lead to unsafe situations and even result in catastrophic consequences.

This paper focuses on the dynamic mixed traffic of CAVs and HDVs, which also takes the heterogeneity among CAVs into account. In addition, the multi-constraint traffic scenario is considered, where it is essential for CAVs to avoid collisions with other CAVs, uncontrollable HDVs, and static obstacles, while also ensuring they do not depart from their lanes. Since traditional RL approaches cannot fully address the issues concerning heterogeneity and safety decisions, we aim to devise a novel approach to resolve these challenges. In detail, we have designed a **Heterogeneous Cluster Policy-sharing Penalized Proximal Policy Optimization (HCP4O)** algorithm and a *hybrid-critic training* framework.

B. Prior Research

We now discuss some relevant prior research on Heterogeneous Agents Reinforcement Learning, Safe Policy Optimization, and Multi-Agent Training Framework.

Heterogeneous Agents Reinforcement Learning: Previous works like [20], [21] primarily focus on homogeneous agents, overlooking the heterogeneity of agents in real-world traffic environments. Specifically, it is assumed that all cooperative CAVs in the system possess identical models and Social Coordination Awareness (SCA), thereby sharing the same policy network parameters [25]. Such an assumption overly simplifies the modeling and computational process, but does not accurately reflect the diverse capabilities and decision-making processes of different vehicles in actual traffic scenarios. To address the issue of heterogeneity, several algorithms have been developed, such as HAPPO, HATRPO [26], and HASAC [27]. Though these methods can eliminate the reliance on parameter sharing among agents, they usually suppose that every agent in the environment is heterogeneous. Therefore, these methods struggle to converge and demand exponentially increasing computational resources as the number of agents grows, especially in large-scale multi-agent environments.

Safe Policy Optimization: Meanwhile, one of the main obstacles to applying RL algorithms in real-world problems is the lack of safety guarantees [28], [29]. For example, an RL agent controlling an autonomous vehicle may obtain high rewards through high-speed driving, but it might also face a great risk of collision. The paradigm to address this issue involves combining traditional RL algorithms with safety checkers [30],

[31], [32] or constraints [33], [34], [35], to ensure or promote safety while optimizing driving policy. The basic idea of safety checkers is first to detect dangerous actions in RL and then adjust the actions of the autonomous vehicle to make driving safer. The constrained reinforcement learning approach, which imposes constraints on the policy to ensure that safe actions are taken during the learning and decision-making process, is not in conflict with the approach that incorporates safety checkers. Specifically, the former corrects unsafe decisions at the control level, while the latter generates safer decisions at the decision-making level.

In this paper, we focus on the safe reinforcement learning approach considering constraints. It aims to develop algorithms that learn policies satisfying safety constraints [22], for which a commonly used modelling framework is the Constrained Markov Decision Processes (CMDPs) [34]. To solve CMDPs, the authors of [36] introduced Constrained Policy Optimization that updates the agent's policy under trust region constraints to maximize agent returns while adhering to agent cost constraints. Whereas, Safe-MARL method remains a relatively new field emerging in recent years. Multi-Agent Constrained Policy Optimization (MACPO) [37] is the first safe model-free MARL algorithm developed based on constrained policy optimization [36]. It guarantees monotonic improvement of rewards while theoretically meeting safety constraints at each iteration. However, MACPO is computationally expensive, as it necessitates computing the Fisher Information Matrix approximation of conjugate gradients. Moreover, it suffers from massive hyperparameters and struggles to scale to multi-constraint traffic scenarios, such as collision avoidance and lane-keeping constraints. To reduce computational time in the most challenging tasks and to facilitate easier implementation, [37] further proposed the first-order model-free safe MARL algorithm, called MAPPO-Lagrangian. However, it still struggles to extend to multi-constraint traffic scenarios.

Multi-Agent Training Framework: The training framework indeed has a significant impact on the performance and effectiveness of RL algorithms [38]. Decentralized Training with Decentralized Execution (DTDE) represents an approach where both training and execution of agents are decentralized. This often leads to issues such as non-stationarity and coordination problems among agents, as each agent independently updates its policy based on its partial observation of the environment, which however can rapidly change as other agents update their policies [39]. While under the framework of Centralized Training with Decentralized Execution (CTDE), the policy update of each agent entirely depends on the guidance of the centralized critic [40]. This approach mitigates the non-stationarity problem by providing a stable training environment where the critic has access to the global state and actions of all agents [41]. However, incorporating safety considerations solely into the centralized evaluation might lead to a mismatch: the global rating does not align with the actual contributions of heterogeneous agents. This mismatch can negatively impact the training effectiveness, resulting in suboptimal outcomes. Therefore, both CTDE and DTDE are unsuitable for environments with heterogeneous agents.

C. Main Contributions

Although **Heterogeneous Agents Reinforcement Learning** methods like [26], [27] can resolve the heterogeneity of agents, they often treat every agent as a heterogeneous type, and hence suffer from low efficiency and poor scalability. In response to the aforementioned challenges, we propose a cluster policy-sharing mechanism suitable for a wide range of heterogeneous mixed traffic scenarios. Considering the model and SCA consistency of same-type CAVs, we treat vehicles of the same type as a cluster, which shares the same policy network parameters within the cluster. Therefore, this mechanism can not only accelerate the training convergence speed but can also significantly reduce the demand for computational resources.

In view of the shortcomings in current **Safe Policy Optimization** research, we address constraints using the penalty function method and propose a first-order safe policy update method. Compared with the previous work [37], our method avoids reliance on second-order information and introduces only one hyperparameter, which is beneficial for the algorithm tuning process. Moreover, by integrating constraints into the objective function through ReLU operators, this approach is more easily scalable to multi-constraint traffic scenarios. Then by combining the proposed cluster policy-sharing mechanism with the safe policy update method, we propose the HCP4O algorithm.

Furthermore, given the shortcomings of current **Multi-Agent Training Frameworks** in heterogeneous agent environments, we design a novel hybrid-critic training framework. Noting that the centralized critic module assesses overall contributions based on global information, we introduce a distributed critic module customized for specific cluster safety to complement global evaluation. Furthermore, the special distributed critic module for heterogeneous clusters promotes behavioral diversity to a certain extent, aiding in balancing the trade-off between common objectives and individualized diversity.

In this work, by randomly combining various micro-traffic scenarios such as straight roads, curves, and intersections, we have created a complex road environment that includes HDVs and static obstacles. Unlike most existing studies that focus only on simple intersections or road scenarios [42], [43], our approach trains agents in large-scale and dynamic settings, aiming to enhance their generalization capabilities. The main contributions of this paper are summarized as follows:

- We introduce a cluster policy-sharing mechanism for heterogeneous CAVs in realistic mixed traffic scenarios. This mechanism allows for policy sharing only among CAVs of the same type during training, which aligns more closely with practical application requirements.
- We propose a first-order safe policy update method, characterized by high sample utilization. We then combine it with the cluster policy-sharing mechanism to propose the HCP4O algorithm.
- We design a novel hybrid-critic training framework based on V2X communication technology. This framework integrates centralized critic and distributed safe critics, thereby enhancing the safety performance of heterogeneous CAVs in various scenarios.

D. Paper Organization

The remainder of the paper is organized as follows. Section II presents the problem formulation, while Section III introduces the methodology for safe policy optimization. This combined with the designed cluster policy-sharing mechanism gives a practice algorithm in Section IV, whereas the hybrid-critic training framework is introduced as well. Extensive comparative experiments are demonstrated in Section V. Finally, some concluding remarks are given in Section VI.

II. PROBLEM FORMULATION

In this section, we first introduce the scenario of cooperative driving among heterogeneous CAVs, and then model this problem as an extended constrained Markov game.

A. Scenario Descriptions of Cooperative Driving Among Heterogeneous CAVs

The future urban traffic scenario is expected to be an environment of mixed driving with heterogeneous CAVs and HDVs. Therefore, we focus on designing the cooperative driving scheme for heterogeneous CAVs in urban environments with dynamic and static obstacles. Supported by V2X services, CAVs are capable of exchanging information and sending service requests to other nearby CAVs, roadside edge computing servers, and remote cloud server, as shown in Fig. 1. To avoid confusion, all HDVs are represented by gray vehicles, as part of the dynamic elements of the environment. In this environment, we use the mixed IDM-Policy from [44] to control HDVs, encompassing both longitudinal and lateral control. Specifically, HDVs' longitudinal behavior is controlled by the Intelligent Driver Model (IDM) [45], and their lateral behavior is modeled by the Minimizing Overall Braking Induced by Lane changes model [46], as detailed in Appendix B. Meanwhile, sudden incidents and obstacles on the road, such as warning triangles, traffic cones, and barricades, are considered static obstacles.

Additionally, the vehicles of different colors in Fig. 1 represent various types of CAVs, including sedans, vans, and trucks. The heterogeneity of these CAVs is primarily reflected in their models and Social Coordination Awareness (SCA). To be specific, the model heterogeneity of CAVs includes mass, tire parameters, as well as their driving characteristics. The parameter configuration details for different types of vehicles are provided in Table IV (See Appendix C). This implies that even if the same action instruction is applied to different types of CAVs, it will result in diverse driving performances. Additionally, the SCA heterogeneity of CAVs is manifested in their varying degrees and scopes of cooperation, e.g., different types of CAVs differ in their cooperative rewards.

In this study, we explore the cooperative driving of heterogeneous CAVs in complex environments. We approach this problem by formulating it as an expanded constrained Markov game, to enable agents to maximize rewards while attempting to satisfy multi-constraints such as collision avoidance and lane-keeping. Through the analysis of interaction data, our objective is to find

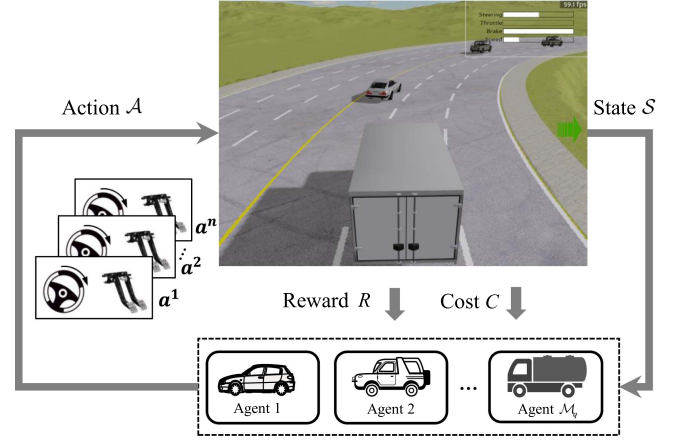


Fig. 2. The interaction process between agents and the environment.

a driving strategy that achieves a balance between efficiency and safety.

B. Reformulation as an Expanded Constrained Markov Game

We formulate the safe cooperative driving problem for q different types of CAVs as an expanded constrained Markov game, denoted by $\langle \mathcal{N}, \mathcal{M}, \mathcal{S}, \mathcal{A}, P, \rho^0, \gamma, R, C \rangle$. Here, $\mathcal{N} = \{1, 2, \dots, n\}$ represents the set of CAVs, and $\mathcal{M} = \{\mathcal{M}_1, \mathcal{M}_2, \dots, \mathcal{M}_q\}$ denotes the set of heterogeneous clusters with $\bigcup_{j=1}^q \mathcal{M}_j = \mathcal{N}$ and $\mathcal{M}_j, j = 1, \dots, q$ denoting the set of CAVs of type j . The function $P: \mathcal{S} \times \mathcal{A} \times \mathcal{S}$ describes the probabilistic transition dynamics, ρ^0 is the initial state distribution, and $\gamma \in [0, 1]$ is the discount factor. We define $\rho_\pi(s) = \sum_{t=0}^{\infty} \gamma^t P(s_t = s)$ to represent the state visitation frequencies under policy π . We denote a trajectory generated by policy π as $\tau = (s_t, a_t, s_{t+1}, \dots)$. The trajectory can be obtained by agents interacting with the environment, as depicted in Fig. 2. Specifically, each agent $i \in \mathcal{N}$ in state s_t selects an action a_t^i based on its policy $\pi^i(a_t^i | s_t)$ at time step t . Upon taking these actions, the agents collectively receive a reward and each agent i incurs a cost. The state of the environment then transitions to s_{t+1} , which is sampled from the probability distribution $P(\cdot | s_t, a_t)$. Next, we will introduce the state space $\mathcal{S}$, action space $\mathcal{A}$, reward function R , and cost function C .

1) *State Space*: The state space is defined as $\mathcal{S} = \prod_{i=1}^n \mathcal{S}^i$. For each agent $i \in \mathcal{N}$, its individual state space $\mathcal{S}^i$ integrates observational information $\mathcal{O}^i$ and navigation information $\mathcal{D}^i$ via V2X communication.

The observational space $\mathcal{O}^i$ of agent i includes the following elements: the steering angle δ^i , heading ϕ^i , current velocity v^i , relative distance to road boundaries e^i , and LiDAR-like point clouds data l^i of surrounding obstacles. This observational space $\mathcal{O}^i$ can effectively represent the main motion characteristics of agent i and the relative distance information of the nearest obstacles around it, which can be expressed as

$$\mathcal{O}^i = [\delta^i, \phi^i, v^i, e^i, l^i]. \quad (1)$$

Additionally, the Cloud Server calculates the navigation route based on the start and end point location information. The

navigation information $\mathcal{D}^i$ comprises the distance information from the current position to the next waypoints d^i and directional information h^i . Therefore, the state space $\mathcal{S}^i$ of agent i can be expressed as

$$\mathcal{S}^i = [\mathcal{O}^i, d^i, h^i]. \quad (2)$$

2) *Action Space*: Let $\mathcal{A} = \prod_{i=1}^n \mathcal{A}^i$ denote the joint action space. This paper uses two actions as input to control the lateral and longitudinal motion of the target vehicle. Therefore, for each agent $i \in \mathcal{N}$, its action is characterized by two dimensions: the steering angle control δ^i and the throttle control p^i . Therefore, the action space of agent i is

$$\mathcal{A}^i = [\delta^i, p^i] \in [-1, 1]^2. \quad (3)$$

Subsequently, these actions are transformed into continuous control commands at a lower level, depending on the cluster to which agent i belongs to. If $i \in \mathcal{M}_j$, then the steering u_s^i , acceleration u_a^i , and brake signal u_b^i are as follows.

$$\begin{aligned} u_s^i &= S_{\max}^j \cdot \delta^i, \\ u_a^i &= F_{\max}^j \cdot \max\{0, p^i\}, \\ u_b^i &= -B_{\max}^j \cdot \min\{0, p^i\}, \end{aligned} \quad (4)$$

where $S_{\max}^j$, $F_{\max}^j$, and $B_{\max}^j$ denote the maximum steering angle, maximum engine force, and maximum brake force of vehicles within cluster $\mathcal{M}_j$, respectively.

3) *Reward Function*: The joint reward function is defined as $R \triangleq \sum_{i=1}^n R_i : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, in which the reward R_i of agent $i \in \mathcal{N}$ is the combination of an individual reward with a cooperative reward. We will detail them in the following.

For each agent $i \in \mathcal{N}$, its individual reward includes both dense and sparse reward.

$$r^i = \underbrace{m_1 \cdot r_{\text{spe}}^i + m_2 \cdot r_{\text{lon}}^i + m_3 \cdot r_{\text{com}}^i}_{\text{Dense reward}} + \underbrace{m_4 \cdot r_{\text{des}}^i}_{\text{Sparse reward}}. \quad (5)$$

The speed reward for agent i is represented as $r_{\text{spe}}^i = v_t^i / V_{\max}^i$ with $V_{\max}^i$ denoting the maximum speed of agent i , which implies that the high-speed driving will result in higher rewards. The longitudinal reward $r_{\text{lon}}^i = y_t^i - y_{t-1}^i$, where y_t^i and y_{t-1}^i denote the longitudinal movement of the agent i in the Frenet coordinates of the current lane between two consecutive time steps. Given the importance of driving comfort in real-world scenarios, we have integrated it into the reward function. The comfort reward for agent i is denoted by r_{com}^i . Specifically, we define the comfort reward based on the following principle: if the present reference route is straight, the yaw angle should be minimized to decrease any unnecessary oscillations. The last term r_{des}^i denotes the sparse reward, which is non-zero only when the target vehicle reaches the destination.

Next, we will design the reward function based on heterogeneous Social Coordination Awareness (SCA). Specially, for each cluster $\mathcal{M}_j$, we introduce the collaborative radius x^j . Here, a larger value of x^j signifies that agents within the cluster $\mathcal{M}_j$ consider a broader range when making decisions, implying a higher level of collaboration. Then we define the heterogeneous cooperative reward for agent $i \in \mathcal{M}_j$ as $\bar{r}_{x^j}^i$, which represents

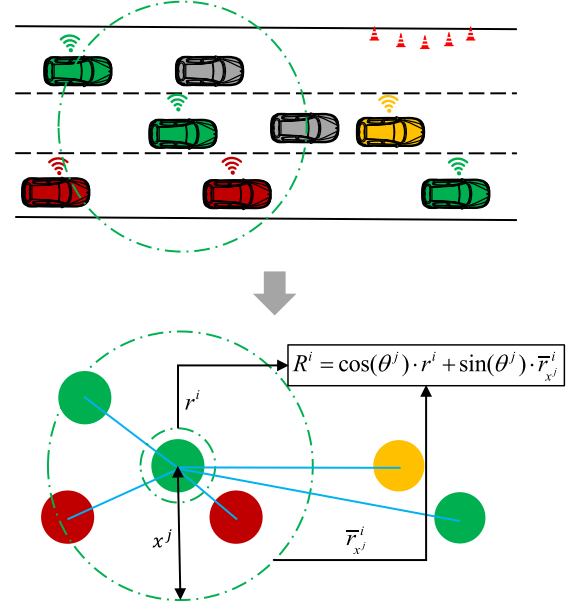


Fig. 3. Description of the reward structure utilizing heterogeneous SCA.

the average reward of CAVs within a radius of x^j meters around agent i , as illustrated in Fig. 3. Therefore, the reward function for agent $i \in \mathcal{M}_j$ is

$$R^i = \cos(\theta^j) \cdot r^i + \sin(\theta^j) \cdot \bar{r}_{x^j}^i. \quad (6)$$

This implies that when the degree of collaboration parameter $\theta^j \in (0, 90)$ of agent i is small, it is more selfish because it tends to optimize its individual rewards while ignoring the rewards of surrounding vehicles.

4) *Cost Function*: The set of cost functions is denoted as $C = \{C_w^i\}_{1 \leq w \leq W^i}$, where each cost function takes the form $C_w^i : \mathcal{S} \times \mathcal{A}^i \rightarrow \mathbb{R}$. Each agent i has W^i cost functions, and the set of corresponding cost-constraining values given by $c = \{c_w^i\}_{1 \leq w \leq W^i}$. Different clusters have constraints corresponding to their characteristics, such as safety and comfort. While within the cluster, the agents have the same cost functions and cost-constraining values. For instance, if agent $i \in \mathcal{M}_j$ has a safety constraint represented by the w -th constraint, its safety cost function includes the costs of vehicle collisions C_{col}^i and driving out of the road C_{out}^i . Therefore, the safety cost function for agent i can be represented as

$$C_w^i = C_{\text{col}}^i + C_{\text{out}}^i,$$

where C_{col}^i and C_{out}^i can be computed as follows:

$$C_{\text{col/out}}^i = \begin{cases} n_{\text{col}}, & \text{if collides,} \\ n_{\text{out}}, & \text{if out of the road,} \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

III. METHODOLOGY FOR SAFE POLICY OPTIMIZATION

In this section, we first introduce the concept of a multi-agent advantage function, based on which we then propose a safe

policy optimization methodology while without considering the clustering structure.

A. Multi-Agent Advantage Decomposition

We consider a fully-cooperative setting where all agents aim to collectively maximize the expected total reward:

$$J(\pi) \triangleq \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right], \quad (8)$$

while also attempt to satisfy individual agent i 's safety constraints,

$$J_w^i(\pi) \triangleq \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t C_w^i(s_t, a_t^i) \right] \leq c_w^i, \quad \forall w = 1, \dots, \mathcal{W}^i. \quad (9)$$

We define the state-action value and the state-value functions in terms of reward as

$$Q_\pi(s, a) \triangleq \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s, a_0 = a \right], \quad (10)$$

$$V_\pi(s) \triangleq \mathbb{E}_{a \sim \pi} [Q_\pi(s, a)]. \quad (11)$$

Notably, the cost value functions $Q_{w,\pi}^i$ and $V_{w,\pi}^i$ resemble the conventional Q_π and V_π . We describe the state-action cost value function and the state cost value function for the w -th cost function of agent i as

$$Q_{w,\pi}^i(s, a^i) \triangleq \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t C_w^i(s_t, a_t^i) \mid s_0 = s, a_0^i = a^i \right], \quad (12)$$

$$V_{w,\pi}^i(s) \triangleq \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t C_w^i(s_t, a_t^i) \mid s_0 = s \right]. \quad (13)$$

Next, we introduce the multi-agent state-action value function. The multi-agent state-action value function for a subset $i_{1:h}$ of n agents and its complement $-i_{1:h}$ is defined as:

$$Q_\pi^{i_{1:h}}(s, a^{i_{1:h}}) \triangleq \mathbb{E} [Q_\pi(s, a^{i_{1:h}}, a^{-i_{1:h}}) \mid a^{-i_{1:h}} \sim \pi^{-i_{1:h}}]. \quad (14)$$

Correspondingly, the multi-agent advantage function for two disjoint subsets $j_{1:k}$ and $i_{1:h}$ is defined as:

$$A_\pi^{i_{1:h}}(s, a^{j_{1:k}}, a^{i_{1:h}}) \triangleq Q_\pi^{j_{1:k}, i_{1:h}}(s, a^{j_{1:k}}, a^{i_{1:h}}) - Q_\pi^{j_{1:k}}(s, a^{j_{1:k}}). \quad (15)$$

For any subset $i_{1:h} \subseteq \mathcal{N}$ with a given joint policy π and state $s \in \mathcal{S}$, the Multi-Agent Advantage Decomposition Lemma [47] is

$$A_\pi^{i_{1:h}}(s, a^{i_{1:h}}) = \sum_{k=1}^h A_\pi^{i_k}(s, a^{i_{1:k-1}}, a^{i_k}). \quad (16)$$

Thus, the joint advantage function $A_\pi^{i_{1:h}}$ can be obtained by summing the individual advantages of each agent. The multi-agent advantage decomposition lemma serves as an effective

Algorithm 1: Safe Policy Optimization.

Input: Stepsize α ; Number of Agents n ; Episodes K ;
Initialize: Actor Networks $\{\theta^{i_h}\}_{i_h=1:n}$;
Output: Actor Networks $\{\theta^{i_h}\}_{i_h=1:n}$;

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1 for  $k = 0, 1, \dots, K-1$  do
2   Collect trajectories by the joint policy  $\pi_\theta$ 
3   Compute  $A_{\pi_{\theta_k}}^{i_h}$  and  $A_{w, \pi_{\theta_k}}^{i_h}, \forall i_h = 1, \dots, n$ 
4   Disrupt the order of Agents:  $i_h \in \mathcal{N}$ 
5   for  $i_h = 1, 2, \dots, n$  do
6     Compute  $\mathcal{L}_{R_{\theta_k}}^{\pi_{\theta_k}}(\theta^{i_h})$  in (22)
7     Compute  $\mathcal{L}_{C_w}^{\pi_{\theta_k}}(\theta^{i_h})$  in (23)
8     Compute objective function:  $\mathcal{L}^{\text{Clip}}(\theta^{i_h}) =$ 
9        $\mathcal{L}_{R_{\theta_k}}^{\pi_{\theta_k}}(\theta^{i_h}) + \kappa \cdot \sum_w \max\{0, \mathcal{L}_{C_w}^{\pi_{\theta_k}}(\theta^{i_h})\}$ 
10    Update Actor Network:
11       $\theta_{k+1}^{i_h} \leftarrow \theta_k^{i_h} - \alpha \nabla \mathcal{L}^{\text{Clip}}(\theta_k^{i_h})$ 
12  end
13 end

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mechanism, enabling agents to enhance their individual learning and collective performance independently. Unlike approaches such as VDN [48] or QMIX [49], this is achieved without the need for assumptions regarding the joint value function's decomposability.

B. Safe Policy Optimization

In this part, we propose a first-order safe policy optimization methodology, as shown in Algorithm 1. In the following, we will explain in details how to design the algorithm.

Given the policy π , denote $\pi^{i_{1:h-1}}$ as a joint policy of agents $i_{1:h-1}$, and π^{i_h} as the policy of agent i_h , respectively. Based on the support of the Multi-Agent Advantage Decomposition Lemma, we can sequentially update the policies of agents. Specifically, when updating π^{i_h} , the policies of agents $i_{1:h-1}$, denoted as $\pi_{k+1}^{i_{1:h-1}}$, have already been updated. We define

$$L(\pi_{k+1}^{i_{1:h-1}}, \pi^{i_h}) \triangleq \mathbb{E} [A_{\pi_k}^{i_h}(\mathbf{s}, \mathbf{a}^{i_{1:h-1}}, \mathbf{a}^{i_h}) \mid \mathbf{s} \sim \rho_\pi, \mathbf{a}^{i_{1:h-1}} \sim \pi_{k+1}^{i_{1:h-1}}, \mathbf{a}^{i_h} \sim \pi^{i_h}]. \quad (17)$$

For any given cost index $w \in \{1, \dots, \mathcal{W}^{i_h}\}$, we define

$$L_w(\pi^{i_h}) = \mathbb{E} [A_{w, \pi_k}^{i_h}(\mathbf{s}, \mathbf{a}^{i_h}) \mid \mathbf{s} \sim \rho_\pi, \mathbf{a}^{i_h} \sim \pi^{i_h}]. \quad (18)$$

We then formulate the unbiased multi-agent constrained policy optimization problem, namely maximizing (8) while satisfying the constraint (9), as follows:

$$\begin{aligned} \pi_{k+1}^{i_h} &= \arg \max_{\pi^{i_h}} \left[L(\pi_{k+1}^{i_{1:h-1}}, \pi^{i_h}) \right], \\ \text{s.t. } &J_w^{i_h}(\pi_k) + L_w(\pi^{i_h}) \leq c_w^{i_h}, \forall w \in \{1, \dots, \mathcal{W}^{i_h}\}. \end{aligned} \quad (19)$$

Iterative search is a commonly employed approach in existing works like [26], [36], [37], which approximate the non-convex Problem (19) by convex optimization via Taylor's formulation on the trust region. In contrast, we integrate constraints with

ReLU operators into the objective function, resulting in an unconstrained problem¹:

$$\begin{aligned} \pi_{k+1}^{i_h} = \arg \min_{\pi^{i_h}} & \left\{ -L \left(\pi_{k+1}^{i_{1:h-1}}, \pi^{i_h} \right) \right. \\ & \left. + \kappa \sum_w \max \{0, L_w(\pi^{i_h}) + J_w^{i_h}(\pi_k) - c_w^{i_h}\} \right\}. \end{aligned} \quad (20)$$

Here, the penalized term is effective when the agent violates a constraint. If all constraints are met, the objective equals the standard policy optimization. However, directly applying existing methods such as Quasi-Newton Methods and the Conjugate Gradient Method to solve Problem (20) poses challenges in scenarios with large state and action spaces.

To address this, we parameterize each agent's policy $\pi_{\theta^{i_h}}$ using a neural network with parameters θ^{i_h} . The joint policies π_θ are parameterized by $\theta = (\theta^1, \dots, \theta^n)$. We use the importance sampling ratio to measure the changes between the new and old policies during the optimization process. Let $r(\theta^{i_h}) = \frac{\pi_{\theta^{i_h}}^{i_h}(\cdot|s)}{\pi_{\theta_k^{i_h}}^{i_h}(\cdot|s)}$ denote the importance sampling ratio. Handling the action distribution through importance sampling, we have

$$\begin{aligned} \theta_{k+1}^{i_h} = \arg \min_{\theta^{i_h}} & \left\{ -r(\theta^{i_h}) L \left(\pi_{\theta_{k+1}^{i_{1:h-1}}}^{i_{1:h-1}}, \pi_{\theta_k^{i_h}}^{i_h} \right) \right. \\ & \left. + \kappa \sum_w \max \left\{ 0, r(\theta^{i_h}) L_w \left(\pi_{\theta_k^{i_h}}^{i_h} \right) + J_w^{i_h}(\pi_{\theta_k}) - c_w^{i_h} \right\} \right\}. \end{aligned} \quad (21)$$

However, directly solving Problem (21) over the unknown future state distribution $s \sim \rho_{\pi_\theta}$ is intractable. We instead use the old policy's corresponding state distribution $s \sim \rho_{\pi_{\theta_k}}$ to minimize the unconstrained objective over collected trajectories. Though it will lead to errors in Problem (21), when the new and old policies are close to each other, the change in the state visitation distribution is minimal, resulting in small error [50]. This makes the approximation reasonable. We then incorporate the constraint via a clipped surrogate objective [51] in the approximated exactly penalized problem to ensure proximal policy iteration. Consequently, $\mathcal{L}_R^{\pi_{\theta_k}}(\theta^{i_h})$ and $\mathcal{L}_{C_w}^{\pi_{\theta_k}}(\theta^{i_h})$ are redefined as

$$\begin{aligned} \mathcal{L}_R^{\pi_{\theta_k}}(\theta^{i_h}) = \mathbb{E} \left\{ -\min \left\{ r(\theta^{i_h}) A_{\pi_{\theta_k}}^{i_h}(s, \mathbf{a}^{i_{1:h-1}}, \mathbf{a}^{i_h}), \right. \right. \\ \left. \left. \text{clip} [r(\theta^{i_h}), 1 - \sigma, 1 + \sigma] A_{\pi_{\theta_k}}^{i_h}(s, \mathbf{a}^{i_{1:h-1}}, \mathbf{a}^{i_h}) \right\} \right. \\ \left. \left| s \sim \rho_{\pi_{\theta_k}}, \mathbf{a}^{i_{1:h-1}} \sim \pi_{\theta_{k+1}^{i_{1:h-1}}}^{i_{1:h-1}}, \mathbf{a}^{i_h} \sim \pi_{\theta_k^{i_h}}^{i_h} \right\}, \right. \end{aligned} \quad (22)$$

$$\begin{aligned} \mathcal{L}_{C_w}^{\pi_{\theta_k}}(\theta^{i_h}) = \mathbb{E} \left\{ \max \left\{ r(\theta^{i_h}) A_{w, \pi_{\theta_k}}^{i_h}(s, \mathbf{a}^{i_h}), \right. \right. \\ \left. \left. \text{clip} [r(\theta^{i_h}), 1 - \sigma, 1 + \sigma] A_{w, \pi_{\theta_k}}^{i_h}(s, \mathbf{a}^{i_h}) \right\} \right\} \end{aligned}$$

¹The constraints are relaxed from (19) to (20). Occasional violations of these relaxed constraints may be beneficial, as detailed in the Section V.

$$+ (J_w^{i_h}(\pi_{\theta_k}) - c_w^{i_h}) | s \sim \rho_{\pi_{\theta_k}}, \mathbf{a}^{i_h} \sim \pi_{\theta_k^{i_h}}^{i_h} \}. \quad (23)$$

The final optimization objective is formulated as

$$\mathcal{L}^{\text{Clip}}(\theta^{i_h}) = \mathcal{L}_R^{\pi_{\theta_k}}(\theta^{i_h}) + \kappa \cdot \sum_w \max \{0, \mathcal{L}_{C_w}^{\pi_{\theta_k}}(\theta^{i_h})\}. \quad (24)$$

Using the clipped surrogate objective technique, we alternately estimate and minimize the unconstrained loss function to solve the challenging constrained problem.

IV. HETEROGENEOUS CLUSTER SAFE ALGORITHM AND HYBRID-CRITIC TRAINING FRAMEWORK

In this section, we first introduce the overall training framework of the proposed algorithm, and then elaborate on the update process in details.

A. Training Framework

We design an algorithm upon an actor-critic structure and show its overall framework in Fig. 4. By incorporating V2X communication technologies, it regards cloud servers as the centralized critics and edge servers as critics focused on cluster security.

1) *Cluster Policy-sharing Actor Module*: In the actor module, we adopt a clustering policy-sharing mechanism. This mechanism allows CAVs of the same type to share policy network parameters, where the policy network parameter of cluster $\mathcal{M}_j$ is denoted by θ^j . This implies that if vehicle $i \in \mathcal{N}$ is a member of cluster $\mathcal{M}_j$, then its policy network parameter is identical to θ^j . This approach is more suitable for real-world applications in mixed traffic environments of connected vehicles, compared to either fully shared policy parameters or completely independent policy parameters. For CAVs, we choose Recurrent Neural Networks (RNNs) to construct their policy networks. This is because RNNs can capture the dynamic characteristics of time series by maintaining internal states while processing and predicting sequence data, displaying a clear advantage over Multi-Layer Perceptrons (MLPs).

Additionally, the actor network is updated based on the guidance of two distinct types of critic networks: *a distributed cost critic network specific to a certain cluster*, and *a centralized critic network*. This approach significantly differs from the traditional CTDE framework. Here, the actor network relies not only on global reward signals but also considers cost evaluations at the cluster level, thereby achieving more fine-grained adjustments and optimization in action selection.

2) *Hybrid-critic Critic Module*: In the critic module, we combine the design philosophies of the Centralized Cloud Critic network and the Distributed Edge safe critic networks. We utilize a centralized critic network based on the global state to evaluate the overall driving behavior of CAVs, which acts as a head coach aiming to achieve reward maximization. Furthermore, in conjunction with the cluster policy-sharing mechanism, we tailor a Distributed Edge safe critic network for each cluster, specifically designed to provide additional policy network update guidance for agents within that cluster when they violate constraints. In

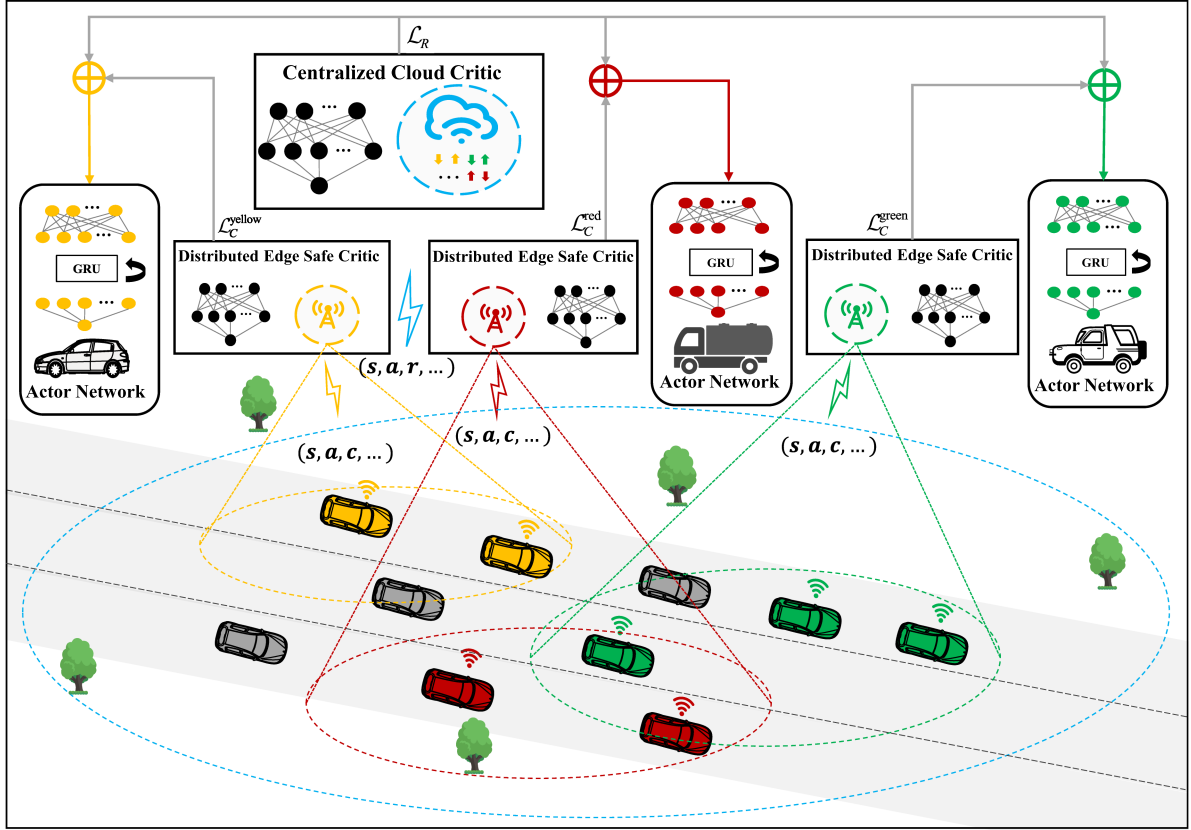


Fig. 4. Illustration of the Hybrid-critic training framework with centralized cloud critic and distributed edge safe critics.

other words, the distributed critic, akin to a personal coach, focuses on encouraging the individual contributions of specific agents within a cluster and restraining unsafe behaviors. It is responsible for evaluating the performance of agents in a specific cluster and provides multi-dimensional guidance for policy learning in conjunction with the centralized critic. Moreover, unique distributed critic networks for different clusters promote behavioral diversity to some extent, aiding in balancing the trade-off between common goals and individualized diversity.

B. Heterogeneous Cluster Policy-Sharing Penalized Proximal Policy Optimization Algorithm

This part focuses on updating the different network modules in the algorithm, as shown in Algorithm 2.

Before updating the neural networks, each agent must interact with the environment to generate trajectories (**Line 3**). From these trajectories, we can derive information necessary for updating the neural networks, such as the cost-related discount value $\hat{C}_t$ and advantage function $A_{w, \pi_{\theta_k}}^{i_h}$, as well as the reward-related discount value $\hat{R}_t$ and advantage function $A_{\pi_{\theta_k}}^{i_h}$ at each timestep (**Lines 4-5**). The following describes the updating process for the three main network modules in detail:

1) *Policy Network Update (Lines 10-12)*: For each agent $i \in M_j$, we first calculate the objective function $\mathcal{L}^{Clip}(\theta_k^j)$ using the advantage function $A_{w, \pi_{\theta_k}}^{i_h}$ and $A_{\pi_{\theta_k}}^{i_h}$, as shown in (24). The

policy network is then updated as follows:

$$\theta_{k+1}^j \leftarrow \theta_k^j - \alpha \nabla \mathcal{L}^{Clip}(\theta_k^j). \quad (25)$$

2) *Distributed Safe Critic Update (Lines 13-16)*: After updating the policy network of their respective clusters, each agent proceeds to update the distributed critic network of the cluster. The Mean Squared Error (MSE) loss function is calculated as

$$L_{\phi_w^j}^{mse} = \frac{1}{bT} \sum_{t=0}^T \sum_{s=1}^b \left(C_{\phi_w^j}(s_t) - \hat{C}_t \right)^2, \forall w = 1, \dots, \mathcal{W}^i. \quad (26)$$

Then the distributed critic network is updated by

$$\phi_{w,k+1}^j = \arg \min_{\phi_w^j} L_{\phi_w^j}^{mse}, \forall w = 1, \dots, \mathcal{W}^i. \quad (27)$$

3) *Centralized Critic Update (Lines 18-21)*: Once all agents have updated their respective policy networks and distributed critic networks, the centralized critic network is updated. The MSE loss function is calculated by

$$L_{\phi}^{mse} = \frac{1}{bT} \sum_{t=0}^T \sum_{s=1}^b \left(V_{\phi}(s_t) - \hat{R}_t \right)^2. \quad (28)$$

Then the centralized critic network is updated as follows.

$$\phi_{k+1} = \arg \min_{\phi} L_{\phi}^{mse}. \quad (29)$$

Algorithm 2: Heterogeneous Cluster Policy-sharing Penalized Proximal Policy Optimization(HCP4O).

Input: Stepsize α ; Number of Agents n ; Episodes K ; Number of clusters q ; batch size b ; steps per episode T ;

Initialize: Actor Networks $\{\theta^j\}_{j=1:q}$; Centralized Critic Network ϕ ; Distributed Critic Network $\{\phi_w^j\}_{w=1:\mathcal{W}^{i_h}}$;

Output: Actor Networks $\{\theta^j\}_{j=1:q}$;

```

1 for  $k = 0, 1, \dots, K - 1$  do
2   for  $i_h = 1, 2, \dots, n$  do
3     Collect trajectories by the policy  $\pi_{\theta^j}^{i_h}$ 
4     Compute  $\hat{R}_t$  and  $A_{\pi_{\theta_k}^{i_h}}$ 
5     Compute  $\hat{C}_t$  and  $A_{w, \pi_{\theta_k}^{i_h}}^{i_h}, \forall w = 1, \dots, \mathcal{W}^{i_h}$ 
6   end
7   Disrupt the order of Agents:  $i_h \in \mathcal{N}$ 
8   for  $i_h = 1, 2, \dots, n$  do
9     if  $i_h \in \mathcal{M}_j$  then
10      Compute  $\mathcal{L}^{Clip}(\theta_k^j)$  in (24)
11      Update Actor Network:
12         $\theta_{k+1}^j \leftarrow \theta_k^j - \alpha \nabla \mathcal{L}^{Clip}(\theta_k^j)$ 
13      Compute loss of Edge Critic Network:
14         $L_{\phi_w^j}^{mse} = \frac{1}{bT} \sum_1^b \sum_{t=0}^T \left( C_{\phi_w^j}(s_t) - \hat{C}_t \right)^2, \forall w$ 
15      Update Edge Critic Network:
16         $\phi_{w,k+1}^j = \arg \min_{\phi_w^j} L_{\phi_w^j}^{mse}, \forall w$ 
17    end
18    Compute loss of Cloud Critic Network:
19       $L_{\phi}^{mse} = \frac{1}{bT} \sum_1^b \sum_{t=0}^T \left( V_{\phi}(s_t) - \hat{R}_t \right)^2$ 
20    Update Cloud Critic Network:
21       $\phi_{k+1} = \arg \min_{\phi} L_{\phi}^{mse}$ 
22  end
23 end
```

V. EXPERIMENTS

In this section, we first present the simulation settings and the methods used for comparison. Then, we demonstrate the superiority of our proposed algorithm and framework through an analysis of overall performance and an ablation study.

Simulation Settings: The experiments are executed on a computing platform equipped with a 2.3 GHz 40-core Intel CPU, 256 GB RAM, and two NVIDIA RTX 4090 GPUs, running Ubuntu 20.04.6 LTS. We conduct our experiments using the lightweight autonomous driving simulator [44], where key hyperparameters are presented in Table III (See Appendix A). By randomly generating complex traffic conditions, we have created 200 diverse maps for each scenario (Sparse, Moderate, and Dense) to simulate various road environments without traffic signals, as shown in Fig. 5. Each scenario includes our heterogeneous CAVs and obstacles, such as HDVs and static obstacles.

Comparative Methods: Our comparative methods are mainly divided into three categories: rule-based methods such as IDM; outstanding MARL methods, including MAPPO, HATRPO, and HAPPO; and safe-MARL methods like MACPO and

TABLE I
SCENARIO CHARACTERISTICS

Scenario	Sparse	Moderate	Dense
Number of CAVs	3	6	9
Total Number of Vehicles	15	25	35
Maximum Steps	1000	1000	1000
Traffic Cones	✓	✓	✓
Straight & Curved Roads	✓	✓	✓
Roundabouts & Intersections	✓	✓	✓
Sedans & Vans & Trucks	✓	✓	✓

MAPPO-L. We also guarantee stable control performance across different random seeds for each method.

A. Overall Performance

1) Comparative Analysis of Algorithm Training across Varied Scenarios: We now conduct training for different traffic congestion scenarios that include three types of vehicles in the environment. We divide traffic conditions into three main scenarios based on their fluidity: Sparse, Moderate, and Dense, where there are approximately 15, 25, and 35 vehicles participating in the traffic flow, respectively. In each scenario, there are 3, 6, and 9 vehicles that are controlled by the actor network as CAVs, while the remaining vehicles are HDVs. The specific scenario characteristics are detailed in Table I. Through such classification, our aim is to construct traffic scenes with varying levels of complexity in order to more accurately evaluate and analyze the performance of various algorithms. The training rewards are shown in Fig. 6, where the solid line represents the average reward across multiple experiments, and the shaded area reflects the standard deviation.

In the Sparse scenario shown in Fig. 6(a), except for the two methods MACPO and MAPPO-L, the other algorithms surpass the reward value of IDM when the training steps reach 3 to 6 million, and gradually stabilize in subsequent training, with a final reward value of about 700. In the Moderate scenario displayed in Fig. 6(b), except for our method that reaches the level of IDM at 4 million training steps, only MAPPO and HAPPO reach the effect of IDM at 6 million steps, and all other methods cannot effectively complete the training. In the Dense scenario demonstrated in Fig. 6(c), our method reaches the effect of IDM even with 9 CAVs at 4 million training steps, and performs better in subsequent training. In contrast, all other algorithms do not reach the reward value of IDM even at 10 million training steps.

Additionally, compared with IDM in the three traffic conditions, our HCP4O algorithm has a stable reward value gap from less than 100in Sparse congestion to 200in Moderate, and over 300in Dense. This change indicates that as the number of vehicles increases, our method demonstrates superior performance compared to IDM. All in all, as seen from Fig. 6, the performance advantage of the HCP4O algorithm becomes increasingly evident with the increase in the number of vehicles in the environment.

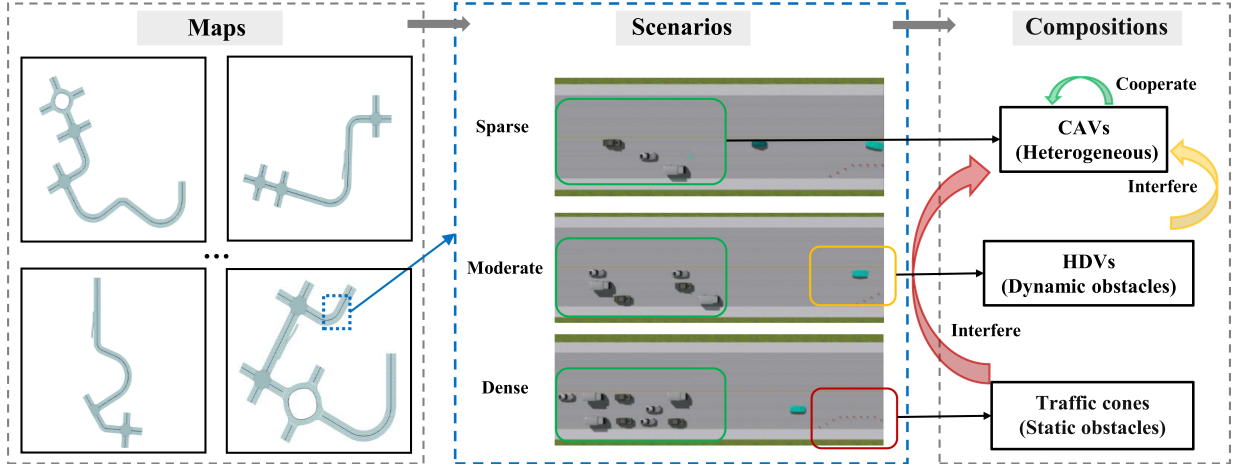


Fig. 5. Description of randomly generated complex traffic conditions.

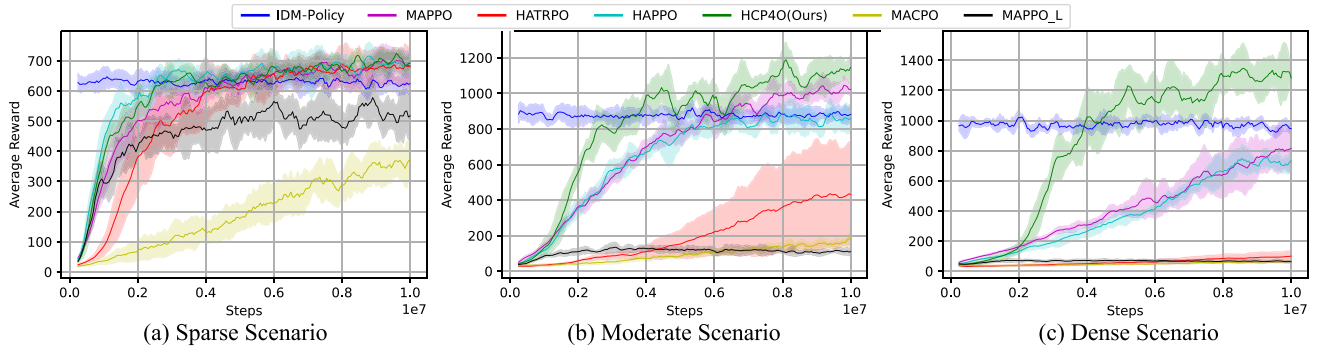


Fig. 6. The training reward curves for various methods in sparse, moderate, and dense scenarios.

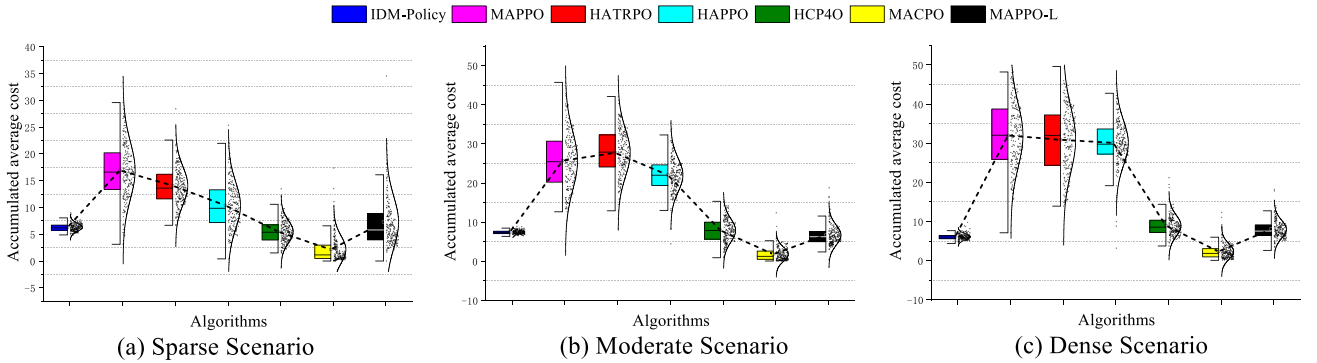


Fig. 7. Cost performance testing of various methods in sparse, moderate, and dense scenarios.

2) Comparative Evaluation of Testing Performance Across Varied Scenarios: In different congestion environments, we test the agents trained by different algorithms. To ensure the statistical validity of the results, each algorithm is tested 100 times in each scenario, with each experiment containing 1000 steps. Figs. 7 and 8 demonstrate the testing costs and rewards, for which the costs and rewards are respectively calculated using the cost function (7) and reward function (6). In addition, we record the average reward (AR), average cost (AC), average

velocity (AV), and average arrival (AA) for all algorithms in Table II with data in the format of means and standard deviations. Here, AV represents the average speed of all vehicles throughout the entire process, while AA indicates the average number of times the vehicles successfully reach the destination within each experiment.

As illustrated in Fig. 7(a), Fig. 8(a), and Table II for the Sparse Scenario, although MARL methods such as MAPPO, HATRPO, and HAPPO achieve high values in terms of reward,

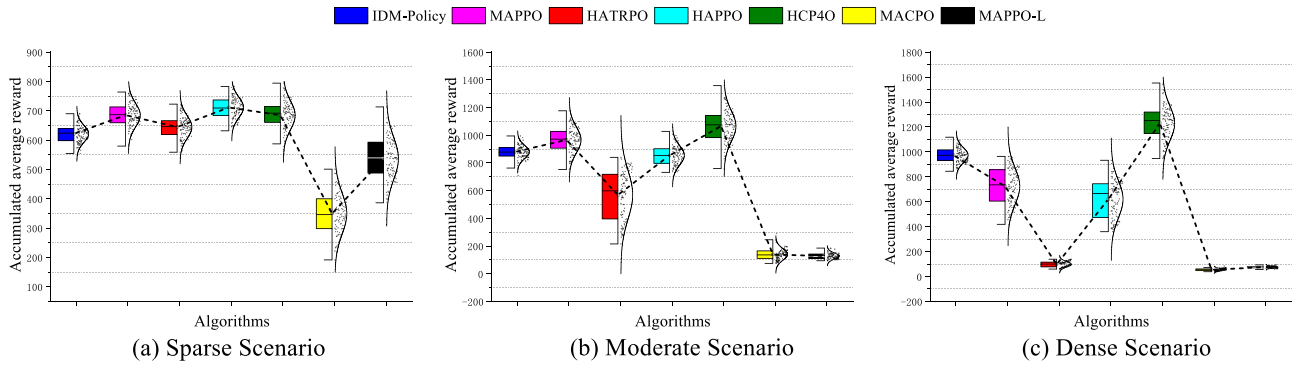


Fig. 8. Reward performance testing of various methods in sparse, moderate, and dense scenarios.

TABLE II
PERFORMANCE COMPARISON ACROSS DIFFERENT SCENARIOS

Scenario	Metric	IDM-Policy	MAPPO	HATRPO	HAPPO	HCP4O(Ours)	MACPO	MAPPO-L
Sparse Scenario	AR↑	619.4 (± 32.2)	685.4 (± 40.4)	644.6 (± 38.5)	711.5 (± 35.2)	684.9 (± 49.4)	349.7 (± 70.2)	537.8 (± 71.0)
	AC↓	6.3 (± 0.7)	16.9 (± 5.4)	14.1 (± 3.5)	10.4 (± 4.4)	5.5 (± 2.1)	2.1 (± 2.6)	7.2 (± 4.8)
	AS↑	5.8 (± 0.2)	6.4 (± 0.3)	5.8 (± 0.5)	6.6 (± 0.4)	6.7 (± 0.3)	0.9 (± 0.2)	5.7 (± 0.2)
	AA↑	7.4 (± 0.3)	7.0 (± 0.4)	8.2 (± 0.4)	6.2 (± 0.3)	7.3 (± 0.4)	0.1 (± 0.1)	1.7 (± 0.2)
Moderate Scenario	AR↑	876.6 (± 49.4)	966.7 (± 93.8)	569.8 (± 176.2)	854.6 (± 70.2)	1039.3 (± 90.8)	139.6 (± 41.8)	127.9 (± 21.8)
	AC↓	7.4 (± 0.5)	25.7 (± 7.4)	27.7 (± 6.1)	22.0 (± 4.3)	7.9 (± 3.0)	1.8 (± 1.8)	6.7 (± 2.6)
	AS↑	4.8 (± 0.6)	5.5 (± 0.5)	4.8 (± 0.5)	4.9 (± 0.2)	5.6 (± 0.5)	0.2 (± 0.1)	3.3 (± 0.3)
	AA↑	5.6 (± 0.2)	4.1 (± 0.5)	5.6 (± 0.4)	5.6 (± 0.5)	7.0 (± 0.4)	0.0 (± 0.0)	1.4 (± 0.2)
Dense Scenario	AR↑	973.4 (± 60.1)	724.4 (± 146.1)	98.1 (± 22.0)	619.8 (± 151.0)	1223.5 (± 122.3)	57.4 (± 9.0)	75.0 (± 7.3)
	AC↓	6.1 (± 0.8)	32.0 (± 8.1)	30.9 (± 8.3)	30.1 (± 5.7)	8.9 (± 2.7)	2.3 (± 2.0)	8.0 (± 2.3)
	AS↑	3.6 (± 0.3)	4.3 (± 0.5)	1.9 (± 0.2)	3.7 (± 0.5)	4.7 (± 0.4)	0.1 (± 0.1)	1.5 (± 0.1)
	AA↑	4.3 (± 0.3)	2.5 (± 0.5)	2.7 (± 0.5)	4.2 (± 0.3)	6.3 (± 0.7)	0.0 (± 0.0)	0.4 (± 0.2)

average speed, and average arrival, their costs are notably higher than those of the rule-based IDM method. On the other hand, safe-MARL methods like MACPO and MAPPO-L exhibit lower costs and higher safety, but they do not attain high values in reward, average speed, and average arrival, suggesting a shortfall in driving performance. This indicates that the learned policy is overly conservative, resulting in almost passive vehicle behavior. This phenomenon is even more pronounced in Moderate Scenario and Dense Scenario, as seen from Fig. 7(b)–(c), Fig. 8(b)–(c), and Table II.

In contrast, our method HCP4O not only achieves the highest driving efficiency in each scenario but also maintains a high level of safety performance comparable to IDM-Policy. Overall, compared to other algorithms, HCP4O more effectively balances safety and efficiency in different scenarios, thereby demonstrating superior decision-making performance.

3) *Analysis of Case Studies in Micro-Scenarios:* To further explore why our algorithm achieves good evaluation results, we conduct a series of experiments to test and analyze the performance of different algorithms across various micro-scenarios. Specifically, we select two algorithms that performed well during both training and testing phases: MAPPO with fully shared parameters and a heterogeneous algorithm HAPPO, as illustrated in Figs. 6 and 8. We compare our proposed HCP4O algorithm separately with MAPPO and HAPPO to demonstrate its advantages in dealing with heterogeneous environments and enhancing safety.

Firstly, we choose a roundabout scenario, where CAVs must enter and exit at the quarter point of the roundabout. In this scenario, MAPPO adopts a shared strategy for all types of CAVs (i.e., trucks and sedans) using the same policy network for training. The results in Fig. 9(a)–(h) show that although this shared strategy enabled trucks to successfully pass the test, sedans would cross the yellow line. This outcome reveals that the shared strategy is not suitable for sedans, mainly due to significant differences in model parameters between sedans and trucks, such as maximum steering angle, acceleration, and speed. Despite normalizing the throttle and steering inputs of all CAVs to the range of $[-1, 1]$, the same action command exhibited different control effects on different vehicle types. In contrast, Fig. 9(i)–(p) shows that our HCP4O algorithm-trained agents are able to effectively control both sedans and trucks to successfully pass the test. This confirms the efficacy of the heterogeneous cluster-sharing mechanism in our algorithm and its ability to train adaptive driving strategies for each vehicle type in heterogeneous environments.

Further, we test the algorithm's performance in a scenario simulating the sudden deceleration of the vehicle ahead. We aim to illustrate why algorithms such as HAPPO and HATRPO exhibit high rewards yet incur high costs, consistent with the data shown in Fig. 7, Fig. 8, and Table II. Specifically, it is seen from Fig. 10(a)–(d) that the strategy trained by the HAPPO algorithm tends to achieve high rewards through high-speed driving, which however ignores the risk of collision and continues to drive at

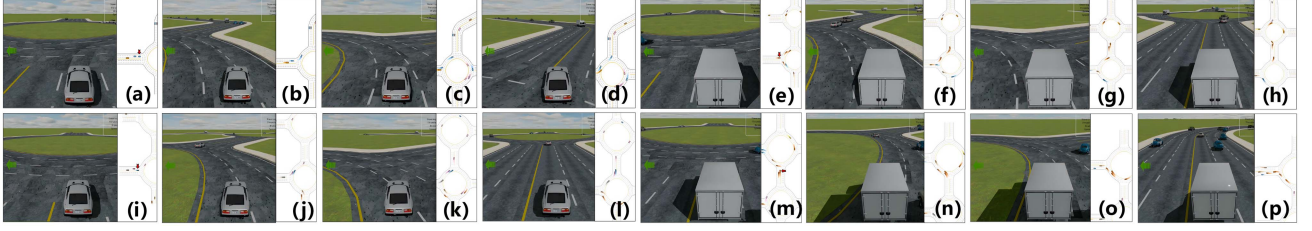


Fig. 9. Heterogeneous performance testing between MAPPO and our method HCP4O in a roundabout scenario: panels (a)–(d) show the scenario of sedans controlled by the MAPPO algorithm failing to navigate through the roundabout; panels (e)–(h) show trucks controlled by the MAPPO algorithm successfully navigating through the roundabout; panels (i)–(l) show sedans controlled by the HCP4O algorithm successfully navigating through the roundabout; and panels (m)–(p) show trucks controlled by the HCP4O algorithm successfully navigating through the roundabout.

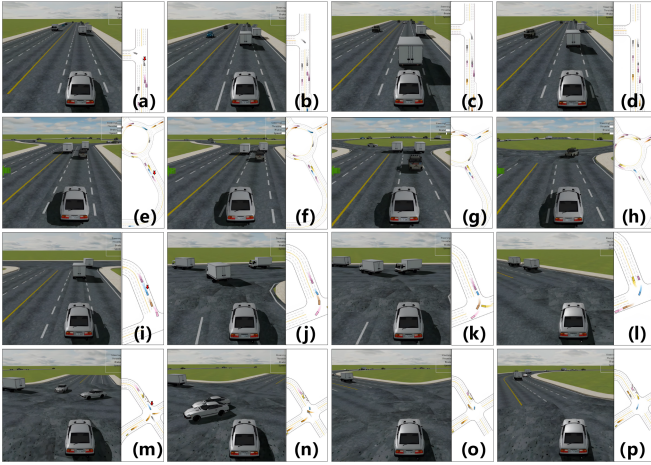


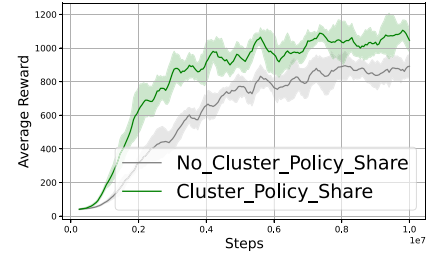
Fig. 10. Safety performance testing between HAPPO and our method HCP4O: panels (a)–(d) depict scenarios where sedans controlled by the HAPPO algorithm are involved in collisions at high speeds; panels (e)–(h) show sedans controlled by the HCP4O algorithm decelerating to avoid collisions; panels (i)–(l) demonstrate sedans controlled by HCP4O navigating through T-intersections, and panels (m)–(p) illustrate sedans controlled by HCP4O safely passing through cross intersections.

high speed even after a collision. On the contrary, facing the sudden deceleration of the vehicle ahead, the strategy trained by the HCP4O algorithm opts to decelerate to reduce the risk of collision and can quickly catch up after the leading vehicle accelerated, as shown in Fig. 10(e)–(h). Overall, this indicates that our HCP4O method is superior in terms of safety compared to state-of-the-art MARL methods.

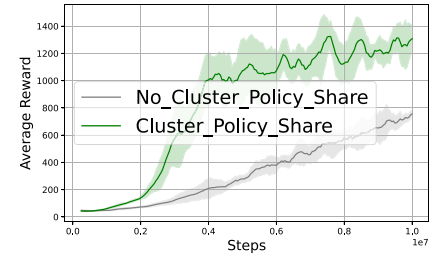
Since most safety collision issues were encountered in complex interaction scenarios, to further demonstrate the superiority of our HCP4O algorithm, we also conduct tests in highly interactive T-junction and crossroad scenarios. Specifically, the results in Fig. 10(i)–(p) show that our CAVs can flexibly evade high-speed vehicles and quickly pass through intersections, highlighting our algorithm’s capability to enhance traffic safety performance. Compared to the IDM-Policy which tends to wait for safety before proceeding, our method demonstrates higher driving efficiency while balancing safety and performance.

B. Ablation Study

1) *Evaluating the Efficacy of the Cluster Policy-Sharing Mechanism:* We carry out experiments within the Moderate and



(a) Moderate Scenario



(b) Dense Scenario

Fig. 11. Ablation study to demonstrate the effectiveness of the cluster policy-sharing mechanism.

Dense Scenario to evaluate the efficacy of the cluster policy-sharing mechanism and demonstrate the results in Fig. 11. As indicated in Fig. 11(a), incorporating the cluster policy-sharing feature in the HCP4O algorithm not only accelerates the training process but also leads to the attainment of higher reward values. Furthermore, Fig. 11(b) showcases the significant role of cluster policy sharing in enhancing sample efficiency, markedly speeding up the training convergence rate. In summary, the integration of the cluster policy-sharing mechanism within the HCP4O algorithm markedly enhances both the efficiency and effectiveness of the training process, as evidenced by faster convergence rates and superior reward outcomes across various scenarios.

2) *Analyzing the Necessity of Distributed Critics:* We conduct tests in the Moderate Scenario to demonstrate the variations in loss values for two different critic training modes and display the empirical results in Fig. 12. Fig. 12(a) shows the change in centralized critic loss value during training, which initially increases from zero to a peak before gradually decreasing. This indicates that the centralized critic initially learns basic driving skills, such as moving forward, before progressively mastering

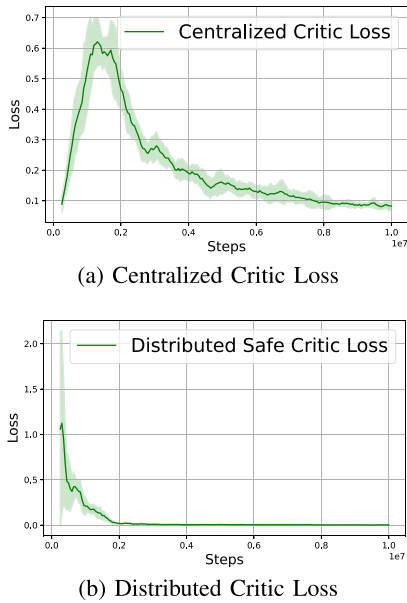


Fig. 12. Losses for the centralized critic and the distributed critic.

more stable and efficient driving techniques. In contrast, Fig. 12(b) illustrates the loss value changes for the distributed critic, which differ significantly from the centralized critic. Since safety evaluation is present from the start, the loss value monotonically decreases from a high value. It suggests that safety evaluation and driving performance evaluation are fundamentally different. Therefore, merely incorporating safety factors into a centralized evaluation system could lead to a mismatch in global scoring and instability in the critic network during training. Thus, it is necessary to introduce a special distributed critic in each CAVs' cluster to assess the safety performance of agents within the cluster. We believe that these specialized distributed critics can be extended to other heterogeneous multi-agent systems, where agent diversity is crucial for the system's effectiveness and robustness in cooperative tasks.

3) *Evaluating the Efficacy of the Hybrid-critic Training Framework*: To validate the superiority of our framework that combines centralized and distributed critics, we conduct a series of ablation experiments focusing on the training framework, with results illustrated in Fig. 13. The blue-colored algorithm represents the HCP4O algorithm without the use of distributed critics, which corresponds to the CTDE framework. The algorithm colored in grey indicates the HCP4O algorithm without the use of a centralized critic, which corresponds to the DTDE framework. It is evident that the HCP4O algorithm, utilizing our innovative hybrid-critic training framework, achieves the fastest convergence speed and the highest reward. Clearly, the DTDE method shows considerable instability in its training process. This is because during the training process, agents only consider their own actions, leading to high variance in the training outcomes and a low average reward. Conversely, while CTDE demonstrates more stability and lower variance during training, it achieves a lower average reward than our proposed training method. Moreover, the results underscore the critical role of the

centralized critic in training multi-agent systems, enabling an effective evaluation of overall performance. Nonetheless, for assessing certain specific metrics, the incorporation of distributed critics is advised to complement the overall assessment.

4) *Analyzing the Computational Efficiency and Complexity Advantages*: We further conduct experiments to illustrate the advantages of our algorithm over other Safe-MARL methods. We perform these experiments in the Moderate Scenario and record the time each algorithm takes to update the neural networks in Fig. 14. MAPPO using a fully shared policy network has an advantage in training time, but it performs poorly in heterogeneous environments. Our method's network update time is comparable to that of HAPPO, though it takes slightly longer due to the need to update the additional distributed critic network. Despite this slight increase in computational cost, introducing the distributed critic network is beneficial as it significantly enhances the algorithm's ability to adapt to dynamic environmental changes. Compared to MACPO and MAPPO-I, our approach reduces the time required per network update by approximately 32% and 38%, respectively. This demonstrates that HCP4O excels in computational efficiency.

5) *Analyzing the Robustness of Hyperparameter κ in Algorithm 1*: The reward function parameters in traditional RL methods are highly sensitive, and improper tuning can lead to risky driving behavior. Notably, in previous related work such as [20], [21], penalties for collisions were simply added to the reward function, making it difficult to tune for favourable results. In contrast, our HCP4O algorithm helps mitigate this issue. To demonstrate it, we further carry out experiments in the Moderate scenario to validate the excellent tuning characteristics of the hyperparameter κ introduced in Algorithm 1. Throughout the experiments, we keep the weights in the reward function constant, as shown in Appendix Table III. We demonstrate the performance of our HCP4O algorithm under different κ values in Fig. 15. It is seen that within a wide range, the algorithm performs effectively, indicating that it does not require precise tuning. This is because we have designed a trainable distributed cost-critic network that can adaptively adjust according to changes in κ . Therefore, this heterogeneous adaptive adjustment capability and robustness are key advantages of our algorithm.

VI. CONCLUSION

This paper aims to address the challenges of cooperative driving and safety decision-making of heterogeneous CAVs in dynamic and mixed complex traffic environments, which traditional RL approaches often struggle to adequately handle. Specifically, we have developed a novel HCP4O algorithm and a hybrid-critic training framework built upon a delicately designed heterogeneous cluster-sharing mechanism. The experimental results demonstrate significant advantages of our approach in both safety and efficiency compared to state-of-the-art RL algorithms and rule-based autonomous driving methods, especially as the number of CAVs increases.

Though our method achieves high efficiency while maintaining low safety costs, the inherent randomness of RL policy implies that absolute vehicle safety cannot be guaranteed.

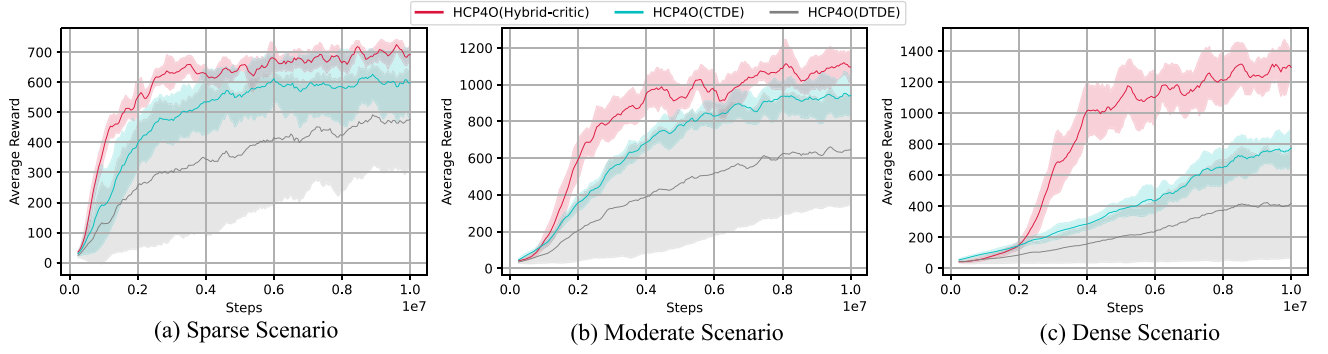


Fig. 13. The reward curves of HCP4O using different frameworks in sparse, moderate, and dense scenarios.

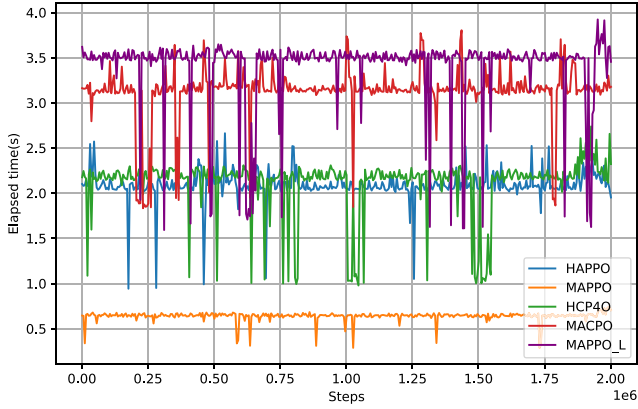


Fig. 14. Illustration of neural network update time comparison.

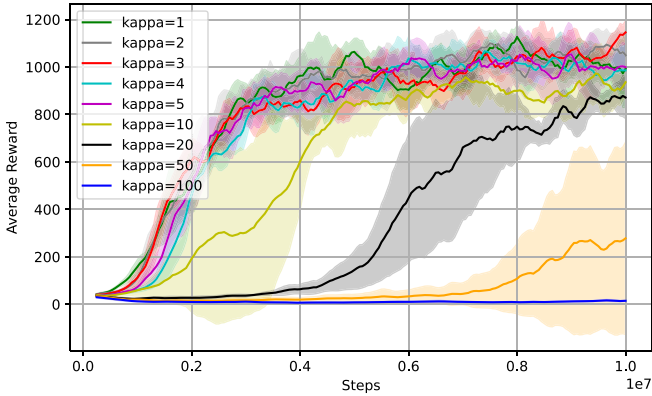


Fig. 15. The performance of HCP4O with different values of κ .

Considering the stringent safety requirements of autonomous driving, it is of interests to further explore the possibility of integrating our method with a safety checker.

APPENDIX A

Since our algorithm employs an on-policy MARL framework, we take the key hyperparameters of the algorithm the same as those of [52]. The details are presented in Table III.

TABLE III
KEY PARAMETERS OF THE ALGORITHM

Train parameters	Value
Number of rollout threads	20
Number of env steps	10e7
Episode length	1000
m_1, m_2, m_3, m_4	0.5, 1, -0.1, 1
n_{col}, n_{out}	2, 1
Destination reward	10
Model parameters	Value
Hidden sizes	128
Activation function	ReLU
Network initialization method	Orthogonal
Output layer gain	0.01
Use GRU	True
Recurrent layers	1
Algorithm parameters	Value
Gamma	0.99
Gae lambda	0.95
Optimizer	Adam
kappa	5
Actor learning rate	5e-4
Critic learning rate	5e-4

APPENDIX B

The actions of CAVs in our problem are decided by the algorithm and will be translated to low-level steering and acceleration signals through a closed-loop PID controller. A Kinematic Bicycle Model determines the vehicle motion [53]:

$$\begin{aligned}
 \dot{x} &= v \cos(\psi + \beta), \\
 \dot{y} &= v \sin(\psi + \beta), \\
 \dot{v} &= a, \\
 \dot{\psi} &= \frac{v}{l} \sin \beta, \\
 \beta &= \tan^{-1}(1/2 \tan \delta),
 \end{aligned}$$

where (x, y) is the vehicle position, v is forward speed, ψ is heading, a is the acceleration command, β is the slip angle at

TABLE IV
VEHICLE SPECIFICATIONS

Metric	Sedan	Van	Truck
Length (m)	4.40	4.50	5.80
Width (m)	1.85	1.86	2.30
Height (m)	1.37	1.85	2.80
Tire Radius (m)	0.39	0.43	0.37
Tire Width (m)	0.30	0.35	0.50
Lateral Tire to Center (m)	0.81	0.75	0.93
Rear Wheelbase (m)	1.21	1.21	1.08
Front Wheelbase (m)	1.28	1.53	1.73
Mass (kg)	1200	1300	1600

the center of gravity, δ is the front wheel angle used as a steering command.

We utilize the mixed IDM-Policy proposed by [44] to control HDVs, encompassing both longitudinal and lateral behavior. Specially, the HDVs' longitudinal acceleration is given by the Intelligent Driver Model (IDM) from [45].

$$\dot{v} = a \left[1 - \left(\frac{v}{v_0} \right)^\delta - \left(\frac{d^*}{d} \right)^2 \right],$$

$$d^* = d_0 + Tv + \frac{v\Delta v}{2\sqrt{ab}},$$

where v is the vehicle velocity, d is the distance to its front vehicle, v_0 is the desired velocity, T is the desired time headway, d_0 is the jam distance, a, b are the maximum acceleration and deceleration respectively, Δv is the velocity exponent. In addition, HDVs' lateral behavior is given by the Minimizing Overall Braking Induced by Lane change model from [46]. According to this model, a vehicle decides to change lanes when

- it is safe (do not cut-in):

$$\tilde{a}_n \geq -b_{\text{safe}},$$

- there is an incentive (for the ego-vehicle and possibly its followers):

$$\underbrace{\tilde{a}_c - a_c}_{\text{ego-vehicle}} + p \left(\underbrace{\tilde{a}_n - a_n}_{\text{new follower}} + \underbrace{\tilde{a}_o - a_o}_{\text{old follower}} \right) \geq \Delta a_{\text{th}},$$

where c is the center vehicle, o is its old follower before the lane change, and n is its new follower after the lane change, $a, \tilde{a}$ are the acceleration of the vehicles before and after the lane change respectively, p is a politeness coefficient, Δa_{th} the acceleration gain required to trigger a lane change, b_{safe} the maximum braking imposed to a vehicle during a cut-in.

APPENDIX C

The parameter configuration details for different types of vehicles are provided in Table IV.

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